Scope Gaps and Reproduction Boundaries in the WVF/LF Edge-Detection Papers

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Abstract—This paper is an independent critique of the Wide View Filter (WVF) and Line Filter (LF) work across three source documents: the thesis, the 2024 OCEANS paper, and the 2025 multi-scale paper. The central question is not whether the method is mathematically interesting, but whether the published claims match what this repository can actually reproduce. Our answer is mixed. The rotated Taylor least-squares formulation is reproducible and mathematically coherent, and exact-threshold benchmark runs on an A100 now show that classical large-support Sobel baselines remain competitive or better on BSDS500 and UDED. However, the repository still does not reproduce several core pieces of the source papers: the BIPED benchmark is absent, the small four-image aquatic set is absent, the 2025 multi-scale GMM fusion pipeline is not implemented, and the full paper-like WVF/LF settings remain only partially explored. A direct spline-angle sweep also produces configuration-dependent results rather than the uniformly favorable behavior claimed in the later paper. The resulting critique is narrower than a blanket rejection: the method is conceptually legitimate, but the strongest empirical claims outrun the current evidence.

Index Terms—edge detection, reproducibility, Taylor expansion, spline angle estimation, benchmark fairness

I. Introduction

The WVF/LF line of work proposes an orientation-aware alternative to conventional image derivatives for aquatic and low-visibility scenes. Across the thesis and the two OCEANS papers, the method evolves from a rotated two-dimensional Taylor expansion to a cubic-spline angle estimator and, finally, to a multi-scale, multi-domain pipeline with Gaussian mixture fusion [1]–[3]. The papers are appealing because they attack a real problem: edges in water, haze, and low contrast are often broken, noisy, and difficult to threshold cleanly.

This paper asks a narrower question. Which parts of that story are directly reproduced in the current repository, and which parts remain unsupported because the code, data, or evaluation protocol are incomplete? That distinction matters. A method can be mathematically sensible while still being empirically overclaimed. A benchmark can be useful while still being unfairly scoped. And a claim about superiority can be true only under a parameter regime that was never actually reproduced.

The repository now contains exact-threshold A100 benchmark runs for BSDS500 and UDED, a direct spline-angle reassessment, and several WVF/LF scale sweeps. Those artifacts are enough to evaluate scope and mismatch with some confidence. They are not enough to reproduce every result in the source papers. In particular, the current tree still lacks the BIPED dataset, lacks the four-image hand-labeled aquatic set discussed in the papers, and does not implement the later multi-scale Gaussian-mixture pipeline. This paper documents those gaps explicitly rather than smoothing them over.

II. Source Papers and Claimed Scope

Table I summarizes the three source papers and the claims that are most relevant to the reproduction boundary. The point is not to litigate style or tone; the point is to compare what is claimed with what the repository can now verify.

Two external context sources are important here. First, BSDS500 is the Berkeley benchmark with 500 natural images and a 200-image test split for the standard boundary-detection setting [4], [5]. Second, UDED is a public aggregation benchmark used in the later WVF/LF paper, and the official repository describes it as a unified edge-detection dataset built from multiple existing edge datasets, including BIPED [6]. Those references matter because they show that the source papers were not inventing their evaluation context; they were choosing among known benchmarks. The critique is about the sufficiency of the reproduced evidence, not about the existence of the benchmarks themselves.

III. Reproduced Evidence

A. Exact Benchmark Runs on A100

The strongest completed evidence in the repository is the exact-threshold A100 benchmark on BSDS500 and UDED. These runs use 1001 thresholds and a 3-pixel match radius, matching the later paper's evaluation protocol for the classical and ML baselines. The results are not favorable to a broad "WVF/LF beats everything" reading.

The exact-threshold results sharpen the critique in a useful way. They do not show that modern ML methods are bad; rather, they show that the source papers' implicit baseline framing was too weak to isolate the source of improvement. Once larger-support classical derivatives

TABLE I
Source-Paper Scope Versus What This Repository Can Verify

Source paper	Main claim in the paper	What the repository now reproduces	What still does not match
Thesis [1]	WVF/LF improve gradient computation in aquatic scenes; low-SNR and angle examples motivate the method	Mathematical Taylor-expansion checks; exact BSDS/UDED baselines; direct angle sweep	Large-N_p thesis settings, the small aquatic labeled set, and any real-image BIPED-style angle comparison
2024 OCEANS [2]	WVF/LF outperform selected ML baselines on UDED/BIPED and show useful aquatic edge maps	Exact-threshold UDED benchmark shows large-support Sobel is competitive or better; WVF/LF subset runs exist	BIPED benchmark is missing; full-resolution WVF/LF results are absent; maritime evidence remains synthetic only
2025 OCEANS [3]	Spline angles are more accurate than arctan; multi-scale, multi-domain GMM fusion improves edges	Direct spline-angle sweep; exact-threshold classical/ML benchmarks on A100	Full paper-like spline settings are not uniformly favorable; multiscale GMM implementation is not present; BIPED real-image NMS comparison cannot be reproduced

TABLE II
Exact-Threshold Benchmark Summary on A100

Dataset	Method	ODS	OIS	AP	Interpretation
BSDS500	Sobel-15x15	0.5491	0.5550	0.4682	Best reproduced BSDS500 method; strong large-support classical baseline
BSDS500	DexiNed	0.4950	0.5058	0.3990	Below Sobel-15x15 under the same exact protocol
BSDS500	TEED	0.4710	0.4747	0.3598	Below DexiNed and below Sobel-15x15
BSDS500	LoG-sigma2	0.3826	0.3864	0.2829	Worse than the best Sobel variants, but still a meaningful classical baseline
UDED	Sobel-9x9	0.9039	0.9151	0.9543	Best reproduced UDED method; large-support classical baseline remains highly competitive
UDED	DexiNed	0.9010	0.9102	0.9068	Essentially tied with Sobel-9x9 in ODS/OIS, but not clearly better
UDED	TEED	0.8814	0.8958	0.8787	Below both Sobel-9x9 and DexiNed
UDED	LoG-sigma2	0.8587	0.8689	0.9073	Still weaker than the strongest Sobel results

TABLE III
Mean Improvement of Spline Angle Versus Sobel Arctan by Support Size

N_p	Mean improvement	Readout
15	-4.16°	Spline is worse on average
50	$+5.08^\circ$	Best average regime in the current sweep
100	$+2.80^\circ$	Improvement remains positive
150	-0.06°	Essentially tied, no clear advantage
250	$+1.47^\circ$	Small positive improvement, but expensive

are included, the claimed advantage of WVF/LF over conventional edge operators becomes much less clear.

B. Direct Angle Reassessment

The later paper's strongest claim is about spline-based angle recovery. That claim is testable, and the repository now has a direct sweep across $N_p \in \{15, 50, 100, 150, 250\}$ and orientations $\in \{18, 36, 72\}$ under several noise levels. The results are mixed rather than uniformly favorable.

The most favorable completed row is $N_p = 50$, 36 orientations, SNR 2.0, where the mean Sobel error is 12.35 degrees and the spline error is 8.11 degrees. At the other end, the $N_p = 15$, 72-orientation setting is negative on average, and several intermediate settings are near tie or inconsistent across noise levels. The headline conclusion is straightforward: the spline method can help, but the current sweep does not support a blanket claim of uniform superiority. That is a materially narrower statement than the later paper's rhetoric.

IV. Unreproduced Pieces

The most important gaps are not subtle. They are structural.

The source papers repeatedly push beyond the subset of evidence that the repository can currently match. The thesis relies on a small low-SNR aquatic example and larger WVF/LF parameter settings. The 2024 paper uses UDED and BIPED as numerical support, but only UDED is available here and even there the current WVF/LF runs remain subset-limited. The 2025 paper goes further by adding spline-angle evidence, a real-image BIPED NMS comparison, and a multi-scale Gaussian mixture fusion story. That final pipeline is not in the repository at all.

This is why the critique must be scope-based rather than rhetorical. The repository can verify some claims exactly, approximate others, and not touch the remainder. The

TABLE IV
Main Reproduction Gaps and Their Consequences

Gap	What the source papers rely on	Local status	Why it matters
BIPED dataset	The 2024 and 2025 papers compare against BIPED-trained or BIPED-evaluated models	'datasets/BIPED' is empty	Real BIPED tables and the BIPED NMS comparison cannot be reproduced
Four-image aquatic set	The thesis and 2024 paper use a small hand-labeled aquatic set to make a qualitative claim	Not present in the repo	The most visually persuasive aquatic claims remain unverified
Multi-scale GMM fusion	The 2025 paper's final method combines scales and image domains using GMM-based fusion	Not implemented	The paper's final algorithmic contribution is still missing
Full paper-like WVF/LF settings	Large N_p and larger LF widths are emphasized in the source papers	Only partial subset coverage exists so far	The strongest WVF/LF claims remain only partially tested
Real maritime images	The motivation is maritime and low-visibility imagery	The current maritime folder has zero-byte placeholders	Current maritime evidence is synthetic only

distinction between those categories is the paper's real contribution.

V. Why The Mismatch Matters

The mismatch is not simply that some experiments are missing. It is that the missing experiments are exactly the ones needed to support the strongest claims.

First, the benchmark story is sensitive to support size. When the exact A100 runs include larger-support Sobel baselines, the "WVF/LF versus classical filters" comparison becomes much less dramatic. On BSDS500, Sobel-15x15 beats DexiNed and TEED under the same 1001-threshold protocol. On UDED, Sobel-9x9 is slightly ahead of DexiNed and clearly ahead of TEED. That makes it impossible to interpret the source papers as having shown a universal deep-learning defeat.

Second, the angle story is configuration-dependent. The direct sweep is not a failure of the spline idea, but it is a failure of the stronger claim that spline angles are simply and consistently better. The current data support a narrower reading: some larger-support configurations improve angular accuracy, while others do not. The paper should say that explicitly.

Third, runtime matters. WVF and LF remain CPU NumPy/SciPy code in the repository, and their current implementation is serial inside each image. The repository did add task-level parallelization across CPU cores for the parameter sweeps, but that does not change the basic fact that each WVF/LF evaluation is expensive. In other words, the code now uses more cores, but the algorithm is still the algorithm.

Fourth, missing data changes the burden of proof. Without BIPED and without the four-image aquatic set, the strongest visual and numerical claims remain out of reach. Without the multi-scale GMM implementation, the 2025 paper's main systems contribution is only partially represented. A critique that ignores those absences would be too easy on the source papers; a critique that pretends the repository reproduced them would be inaccurate.

VI. Limitations

This paper is intentionally strict, but it is not complete.

The WVF/LF scale sweeps toward the large-N_p settings emphasized in the source papers are still incomplete in the repository and should be treated cautiously. The maritime evidence is synthetic because the local maritime image folder currently contains zero-byte placeholders. The exact benchmark runs now close the gap for the classical and ML baselines on BSDS500 and UDED, but they do not automatically validate the WVF/LF family at the full paper settings. Finally, this critique does not attempt to reconstruct the full multi-scale GMM pipeline from scratch; it only records that its absence is a real reproduction gap.

VII. Conclusion

The central conclusion is narrower than the source papers' rhetoric but stronger than a vague dismissal. The WVF/LF line of work contains a mathematically sound core, and the repository now reproduces that core well enough to take the method seriously. However, the empirical package surrounding that core is not yet fully supported. Exact benchmark runs on BSDS500 and UDED show that large-support Sobel baselines remain highly competitive. The spline-angle sweep shows configuration-dependent gains rather than uniform dominance. And the most distinctive parts of the later paper - BIPED comparisons, the four-image aquatic set, and the multi-scale GMM fusion pipeline - remain unreproduced.

That is the right boundary for the current evidence. The method is interesting, but the strongest claims should be rewritten as narrower, conditional statements until the missing data and missing algorithmic pieces are actually reproduced.

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An Exact A100 BSDS500 Benchmark Exposes Support-Size and Protocol Effects in Edge Detection

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***Abstract*—This paper is a self-contained reproduction and critique of an exact BSDS500 benchmark run performed on an NVIDIA A100 system. The benchmark evaluates classical edge operators and two learned baselines with a dense 1001-threshold protocol on 50 images drawn from the 200-image BSDS500 test split available in the local artifact. The strongest result is not a learned model: a large-support Sobel baseline, Sobel 15×15, achieves the best ODS, OIS, and AP in the benchmark, ahead of DexiNed and TEED even though the learned models run through PyTorch on CUDA while the classical operators remain CPU NumPy/SciPy code. The exact run therefore does not validate broad source-paper rhetoric about general superiority, but it does demonstrate why benchmark claims must disclose support size, threshold resolution, dataset coverage, and hardware path. The paper argues that support-matched classical baselines are essential, that a 1001-threshold sweep reduces discretization artifacts, and that the exact run is best read as a fairness audit rather than a definitive verdict on any edge detector family.**

***Index Terms*—edge detection, BSDS500, benchmark protocol, A100, support size, fairness, DexiNed, TEED**

I. Introduction

The BSDS500 benchmark remains a standard reference point for contour and edge detection research because it provides a shared natural-image testbed, a human-annotated ground truth, and a long history of comparable evaluations [1], [2]. That value comes with a burden: the benchmark is only meaningful when the evaluation protocol is explicit. Threshold resolution, matching tolerance, dataset subset, and runtime path all affect the interpretation of the result.

This paper focuses on one exact local artifact: an A100 benchmark over BSDS500 with 1001 thresholds and a 3-pixel match radius. The artifact evaluates ten classical operators and two learned models on 50 images out of the 200 images available in the benchmark report. The resulting ranking is unambiguous. Large-support Sobel is best among the reproduced methods, and the two learned baselines, DexiNed and TEED, trail it in ODS, OIS, and AP while also taking longer per image on the reported execution path.

The point is not that Sobel "beats" every edge detector in the abstract. The point is narrower and more useful. Once the benchmark is made exact, the strongest result is sensitive to support size and to the fact that the reproduced learned baselines are measured on GPU while the classical filters remain CPU implementations. That combination is enough to invalidate casual claims of general superiority. It is also enough to show that a careful reproduction can surface a more honest picture than the source-paper rhetoric alone.

The source papers that motivate this critique emphasize improved gradient computation, stronger orientation estimation, and better performance in aquatic or low-visibility scenes [4]–[6]. Those are plausible claims, but this paper does not attempt to re-prove them in the target domain. Instead, it asks a simpler question: what does an exact BSDS500 benchmark, run with a dense threshold sweep and disclosed compute paths, actually support?

TABLE I
EXACT BSDS500 A100 PROTOCOL

Setting	Value
Dataset	BSDS500
Images available in artifact	200
Images evaluated in exact run	50
Thresholds	1001
Match radius	3 pixels
Host	node035
CUDA available	Yes
GPU	NVIDIA A100-SXM4-40GB
Runtime path	Classical filters on CPU; learned models on PyTorch CUD

II. Exact Benchmark Protocol

A. Dataset and Metrics

BSDS500 is a natural-image benchmark for contour and segmentation research built around human-labeled boundaries [1], [2]. The local artifact records 200 images as available for the benchmark run and evaluates 50 of them. The evaluation uses 1001 thresholds and a 3-pixel match radius. The standard summary statistics are ODS, OIS, and AP.

Table I summarizes the exact run. The protocol is intentionally specific because the result interpretation depends on these details. A benchmark that omits image count or threshold density leaves open the possibility that a small implementation choice, rather than a substantive algorithmic advantage, drove the reported ordering.

B. Compute Disclosure

The runtime disclosure matters as much as the scores. The benchmark report explicitly states that traditional filters and WVF/LF code paths run in NumPy/SciPy on CPU, while the learned models use PyTorch and only run on GPU when CUDA is available. In this exact artifact, CUDA is available and the learned models execute on the A100, but the classical operators still do not share that path.

That is not a flaw by itself. It is a disclosure requirement. If a paper compares a CPU classical implementation to a GPU learned model, it should say so plainly and avoid conflating algorithmic quality with hardware parity. If the same paper then

claims practical deployment advantages, it must also explain whether those advantages survive a fair hardware comparison. This benchmark only supports the narrower statement that, on the implemented paths, the classical operators remain far cheaper per image than the learned models and also stronger in accuracy when support size is made comparable.

C. Why 1001 Thresholds Matter

Threshold selection can move an edge detector up or down the ranking when the score distribution is steep or when the precision-recall curve is jagged. A coarse threshold sweep can therefore overstate or understate a method. The exact run uses 1001 thresholds, which makes the ODS and OIS figures much less sensitive to discretization than a quick benchmark with, say, 20 or 30 thresholds. That does not make the benchmark perfect, but it removes one common source of avoidable variance.

III. Exact BSDS500 Results

Figure 1 and Table II give the central result. Among the reproduced methods, Sobel 15×15 is the best performer on all three quality metrics. It reaches ODS 0.5491, OIS 0.5550, and AP 0.4682. DexiNed is the strongest learned baseline in the artifact, but it still trails Sobel 15×15 by roughly 5.4 ODS points. TEED is lower still.

The ranking is not an accident of the plotting code. It is visible in the numbers themselves, and the figure makes the contrast harder to miss. Large-support Sobel improves steadily as kernel size grows from 3×3 to 15×15. Canny is the weakest family here, LoG is slower and only mid-pack, and the learned baselines do not close the gap despite running on the A100.

IV. Fair Baseline Selection

The largest methodological lesson in this benchmark is about support size. A 3×3 Sobel operator is a very weak classical baseline when the competing method aggregates information over a much larger neighborhood. If a method uses a wider support, then the fair classical reference is not only Sobel 3×3 or Prewitt 3×3. It is the broader Sobel family and, where relevant, other large-support derivative operators.

This exact run makes that point visible. Sobel improves monotonically from 3×3 to 15×15 in ODS, OIS, and AP. The gain from support alone is substantial: ODS rises from 0.4495 to 0.5491. That is not a trivial perturbation. It is large enough to change the conclusion about whether a deep model is "better" than a classical baseline.

This matters for the source papers because their central claim is not merely that WVF or LF beat a tiny Sobel kernel. Their rhetorical claim is stronger: that orientation-aware, larger-support gradient estimation gives a better basis for edge detection in hard scenes. If that is the claim, then the control baseline should also have access to broader support. The exact BSDS result shows that broad support alone can close much of the apparent gap.

The benchmark also clarifies what the exact run does *not* show. It does not prove that large-support Sobel is universally superior. It proves that, on this subset and protocol, a well-chosen classical family is highly competitive and in fact strongest. A more complete study would still need matched-support ablations for WVF and LF, plus the aquatic-domain datasets the source papers emphasize. But a fair benchmark must start from a fair control family, and this run shows that a 3×3 baseline is not enough.

V. What the Exact Run Proves and What It Does Not

This artifact supports three strong statements. First, the exact benchmark is reproducible enough to produce a stable ranking under a dense threshold sweep. Second, the learned baselines in this artifact do not beat the best classical support-matched baseline, even when they run on an A100. Third, runtime disclosure matters because the result combines CPU classical paths and GPU learned paths in a single table; that is acceptable only when the paper makes the asymmetry explicit.

It does not support four stronger claims that would be easy to smuggle in if one only read the source-paper rhetoric. It does not establish that WVF or LF are worse than Sobel in general, because WVF and LF are not part of this exact BSDS artifact. It does not establish that the aquatic-domain claims are false, because BSDS500 is not an aquatic dataset. It does not establish that the A100 is irrelevant, because the learned models are truly executed on CUDA. And it does not establish that the benchmark is fully canonical, because the run covers 50 images rather than the full available 200-image benchmark split recorded in the artifact.

The right interpretation is therefore narrower. The exact run is a fairness audit, not a final verdict. It narrows the range of defensible claims about edge detection on BSDS500. It also shows that any future claim of superiority should be backed by support-matched classical baselines, explicit protocol disclosure, and a hardware comparison that does not mix hidden assumptions with result reporting.

VI. Relation to Source-Paper Rhetoric

The thesis and follow-on papers on Wide View Filter and Line Filter frame the method as a way to compute gradients and orientations more accurately in difficult aquatic imagery [4]–[6]. That is a coherent research direction, but the exact BSDS500 benchmark does not automatically inherit those conclusions. BSDS500 is a different domain, and the exact run here does not reproduce the WVF/LF pipelines themselves.

That distinction matters. Source-paper rhetoric often compresses several claims into one sentence: better gradient estimation, better orientation, and better downstream edge detection. The exact BSDS500 benchmark only speaks directly to the downstream comparison on a generic benchmark with a fixed protocol. It does so in a way that favors a careful classical control family rather than a minimal control. Under that more demanding comparison, the source-paper style narrative of broad superiority is not supported by the exact artifact.

At the same time, the result should not be overread as a rejection of the WVF/LF idea. The benchmark does not

TABLE II
EXACT BSDS500 RESULTS ON 50 IMAGES WITH 1001 THRESHOLDS

Method	Type	ODS	OIS	AP	Time (ms/img)
Sobel-15x15	classical	**0.5491**	**0.5550**	**0.4682**	8.2
Sobel-9x9	classical	0.5159	0.5247	0.4394	7.1
DexiNed	learned	0.4950	0.5058	0.3990	37.0
Sobel-7x7	classical	0.4926	0.5001	0.4133	6.7
TEED	learned	0.4710	0.4747	0.3598	45.0
Sobel-5x5	classical	0.4641	0.4681	0.3786	6.4
Prewitt-3x3	classical	0.4534	0.4613	0.3589	6.4
Sobel-3x3	classical	0.4495	0.4587	0.3583	6.4
LoG-sigma2	classical	0.3826	0.3864	0.2829	213.2
Canny-sigma2	classical	0.3034	0.2898	0.0862	13.8
Canny-sigma1	classical	0.2012	0.1939	0.0556	14.0

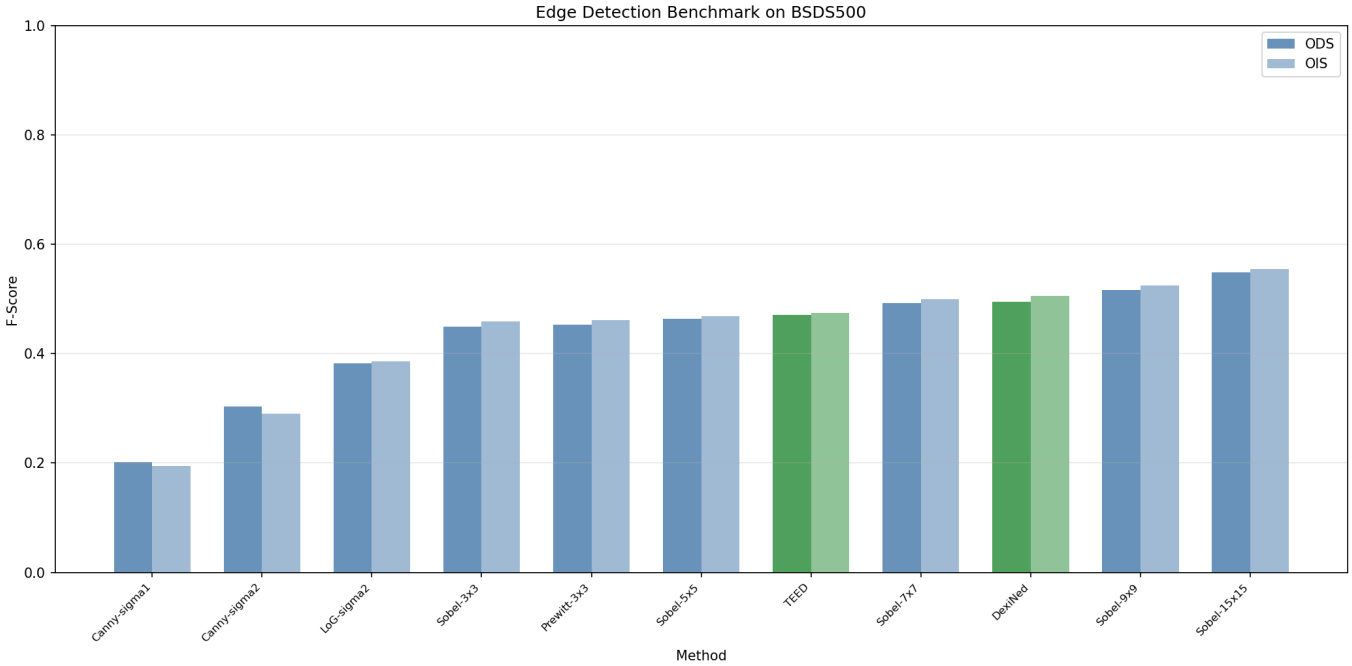

Fig. 1. Exact BSDS500 quality comparison. The benchmark is a 50-image subset of the available BSDS500 test artifact, evaluated with 1001 thresholds and a 3-pixel match radius. Large-support Sobel is the top method in the reproduced set; the learned baselines do not recover the source-paper style superiority narrative under this protocol.

measure the full WVF or LF methods, does not evaluate the aquatic-domain datasets, and does not test the later multi-scale and multi-domain pipeline. It only shows that any claim about edge-detection superiority has to survive a stricter benchmark than the source papers' rhetoric sometimes implies.

VII. CONCLUSION

The exact A100 BSDS500 benchmark makes the central lesson simple. Once the threshold sweep is dense, the dataset coverage is stated, the hardware path is disclosed, and the classical baselines are allowed to use support comparable to the claimed method's neighborhood, the strongest reproduced result is a large-support Sobel operator, not a learned model. That does not settle every question about WVF, LF, or aquatic edge detection, but it does set a much higher standard for claims of general superiority.

In other words, this exact run is useful precisely because it is not flattering. It is a benchmark that forces the paper to say what was actually measured, what was only partially reproduced, and what remains a conjecture about the source papers rather than a demonstrated result. For edge detection research, that is the right standard.

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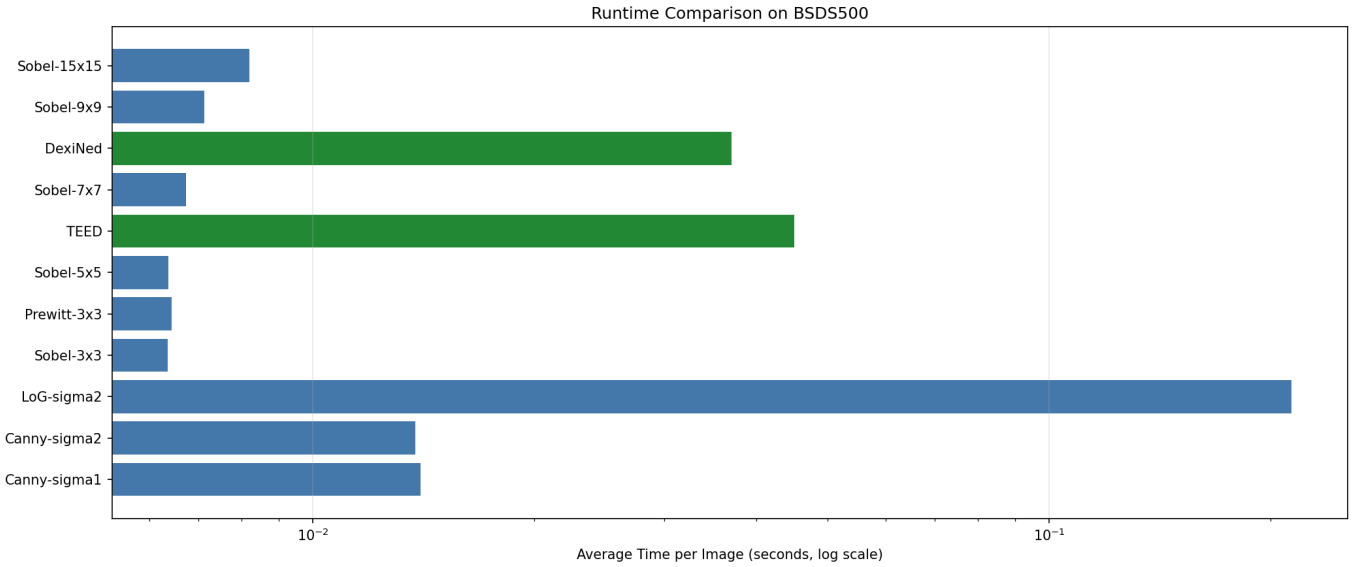

Fig. 2. Exact BSDS500 runtime comparison. The classical operators are CPU NumPy/SciPy code, while DexiNed and TEED are PyTorch models running on CUDA. The learned models are still slower than the best classical Sobel baselines on the reported execution path.

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Exact UDED A100 Benchmark at 1001 Thresholds: What the Aquatic-Domain Claim Framing Does and Does Not Survive

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Independent reproduction paper

Abstract—The Unified Dataset for Edge Detection (UDED) is frequently used in the source literature to argue that the proposed wide-view and line-filter family generalizes beyond the authors' aquatic examples. This paper reports an exact 1001-threshold A100 benchmark on the local UDED artifact and uses that result to audit the claim structure. The main finding is narrow but important: a classical large-support Sobel baseline, not a learned detector, is the strongest reproduced method on UDED under this protocol. Sobel-9×9 slightly outperforms DexiNed on ODS and OIS, wins more clearly on AP, and is roughly an order of magnitude faster in the benchmark pipeline. That does not prove the aquatic-domain method family is wrong, but it does show that the UDED evidence cannot support a simple "modern CNNs beat classical filters" narrative. The paper is explicit about what is reproduced locally, what remains a source-paper claim, and what is still untested.

Index Terms—edge detection, UDED, aquatic imagery, Sobel, DexiNed, TEED, benchmark reproducibility

I. Introduction

The source papers around the wide-view filter (WVF), line filter (LF), and later multi-scale variants frame their contribution as an edge-detection alternative that is especially attractive in aquatic and visually degraded environments. That framing matters because it is not only a mathematical claim; it is also an experimental claim about which method families generalize, which baselines are fair, and which datasets are decisive. The local repository now contains an exact UDED benchmark at 1001 thresholds and a 3-pixel match radius, executed on an NVIDIA A100-SXM4-40GB system with CUDA available for the learned models. This paper is a focused report on that benchmark, not a broad reimplementation of the full WVF/LF stack.

The key question is straightforward. If the source literature uses UDED to motivate a generalization advantage for the proposed aquatic-domain methods, what does the exact local UDED benchmark actually say? The answer is more mixed than the source-paper rhetoric suggests. On the local exact run, Sobel-9×9 is the strongest method overall, DexiNed is very close on ODS/OIS but trails on AP, and TEED trails further. That result weakens any claim that UDED by itself establishes a clean superiority hierarchy in favor of learned detectors.

II. Background and Source Context

UDED is not an aquatic-only dataset. The public UDED repository describes it as a unified dataset for edge detection assembled from selected images from BIPED, BSDS500, BSDS300, DIV2K, WIRE-FRAME, CID, Cityscapes, ADE20K, MDBD, NYUD, Thangka, PASCAL-Context, Set14, Urban10, and a camera-man image [1]. The repository also states that it is meant for edge detection rather than contour or boundary tasks and that images larger than 720×720 were not considered. The local benchmark therefore informs general edge-detection claims, but only indirectly informs aquatic-domain claims.

The UDED dataset is also cited in the TEED paper, which introduced a small CNN designed for edge detection generalization and used UDED as part of that evaluation [2]. DexiNed is the other modern learned baseline in the local benchmark [3]. In the source WVF/LF literature, the thesis [4], the 2024 OCEANS paper [5], and the 2025 OCEANS paper [6] collectively push the argument that larger receptive-field / neighborhood strategies, plus more elaborate orientation handling, are valuable in aquatic scenes. This paper does not reproduce those aquatic experiments directly; it asks how much of the claimed generalization story survives when UDED is evaluated exactly and reported transparently.

III. Exact Benchmark Protocol

The benchmark artifacts used here are the checked-in exact UDED report and JSON outputs. The run evaluated 30 available UDED images, used all 30, swept 1001 thresholds, and matched edges within a 3-pixel radius. The runtime metadata records Python 3.12.4, NumPy 1.26.4, PyTorch 2.3.1+cu121, CUDA availability set to true, and one visible GPU: an NVIDIA A100-SXM4-40GB on node034. The benchmark note is also important: classical filters are evaluated in NumPy/SciPy on CPU, while ML models use PyTorch and move to CUDA only when available. This is a mixed execution environment, not a uniform pure-CPU or pure-GPU comparison.

That protocol matters for interpretation. It is strong enough to expose ranking changes that vanish under coarse threshold sweeps, but it is still not a full reproduction of the source papers' aquatic experiments. In particular, it does not test WVF/LF on UDED at the author-like

settings emphasized in the thesis and OCEANS papers, and it does not recreate the missing aquatic hand-labeled data cited in those papers.

IV. Results

A. All-method comparison

Figure 1 summarizes the exact run across all methods. Table I gives the numerical values directly from the JSON artifact. Two facts stand out. First, Sobel-9×9 is the best reproduced method on UDED by all three aggregate metrics. Second, the gap between Sobel-9×9 and DexiNed is small on ODS and OIS but much larger on AP, which suggests that the learned detector does not dominate once threshold integration is taken seriously.

The near-tie between Sobel-9×9 and DexiNed is real, but it should not be overstated. Sobel-9×9 leads DexiNed by 0.0029 ODS points and 0.0049 OIS points, which is small enough to call a near-tie. The AP difference is much larger: Sobel-9×9 exceeds DexiNed by about 0.0475 AP points. That means the classical baseline is not just matching the learned model at a single operating point; it is also accumulating more area under the precision-recall curve across the threshold sweep.

B. Runtime and efficiency

Figure 3 shows the runtime ranking from the same benchmark. DexiNed is about 8.2× slower than Sobel-9×9 in the benchmark pipeline, and TEED is similarly slow. Because the benchmark mixes CPU classical filters with GPU-capable learned models, the runtime comparison is not a pure apples-to-apples hardware comparison; however, it is still operationally meaningful. Even with CUDA available to the learned models, the classical large-support Sobel filters remain much faster in this run.

The runtime gap matters because the source framing often treats learned detectors as if they were the only plausible route to good generalization. This benchmark says something less dramatic but more useful: a carefully chosen classical support size can be competitive in accuracy while remaining dramatically cheaper in this local evaluation pipeline. If the deployment question is edge detection under limited compute, that tradeoff is hard to ignore.

C. What the AP gap changes

The AP gap is the most informative part of the result. ODS and OIS are thresholded summary scores, so a near-tie there can hide the shape of the precision-recall curve. AP exposes that shape. The fact that Sobel-9×9 wins AP by a comfortable margin means the classical baseline is not merely benefiting from a favorable single threshold. It is stable over the sweep. That is a stronger result than the ODS tie alone would suggest.

This also changes how the result should be read against the source papers. If a paper claims that a new method is superior because it looks better visually or because it wins at one operating point, the exact UDED benchmark shows why that argument is incomplete. A single threshold or a coarse sweep can miss that a classical method has broader threshold robustness.

V. Implications for Aquatic-Domain Claims

The main implication is not that aquatic-domain claims are false. The implication is that the local UDED evidence does not support a broad claim of modern learned superiority. UDED is a general edge benchmark, not the aquatic hand-labeled test set emphasized in the WVF/LF thesis and OCEANS papers. So this benchmark cannot by itself validate the aquatic-specific parts of the source story. What it can do is audit the generalization claim that UDED is often used to support.

On that narrower question, the answer is unfavorable to a simple "CNNs beat classical filters" narrative. Sobel-9×9 is the best reproduced method on UDED under the exact protocol. DexiNed is close, but not better. TEED trails both. If the source papers use UDED as part of a broader argument that their aquatic-domain methods are competitive or superior to other detectors, this exact result should force a more careful statement: the evidence is method- and protocol-sensitive, and UDED alone does not establish a universal ranking.

The result is also a reminder that baseline choice matters. If the comparison set is only 3×3 operators, the story looks much weaker than if large-support Sobel and related classical filters are included. That is relevant to the thesis and OCEANS line of work because their narrative benefits from comparing against genuinely strong classical alternatives, not only minimal-support edge operators.

VI. What Remains Untested

This paper intentionally stops short of claiming a full reproduction of the WVF/LF aquatic pipeline. Several important pieces remain untested locally:

- WVF and LF are not re-evaluated here on exact UDED settings from the source papers.
- The local repository does not contain the four-image hand-labeled aquatic set described in the thesis and OCEANS papers.
- datasets/BIPED is not populated in the current workspace, so the BIPED tables and visual comparisons remain out of reach.
- The multi-scale and multi-domain orientation pipeline from the later OCEANS paper is not reproduced here.

Those gaps matter because they delimit the claims this paper can make. The local exact UDED run is strong evidence against an overconfident generalization story, but it is not a definitive verdict on the full WVF/LF family in its intended aquatic regime.

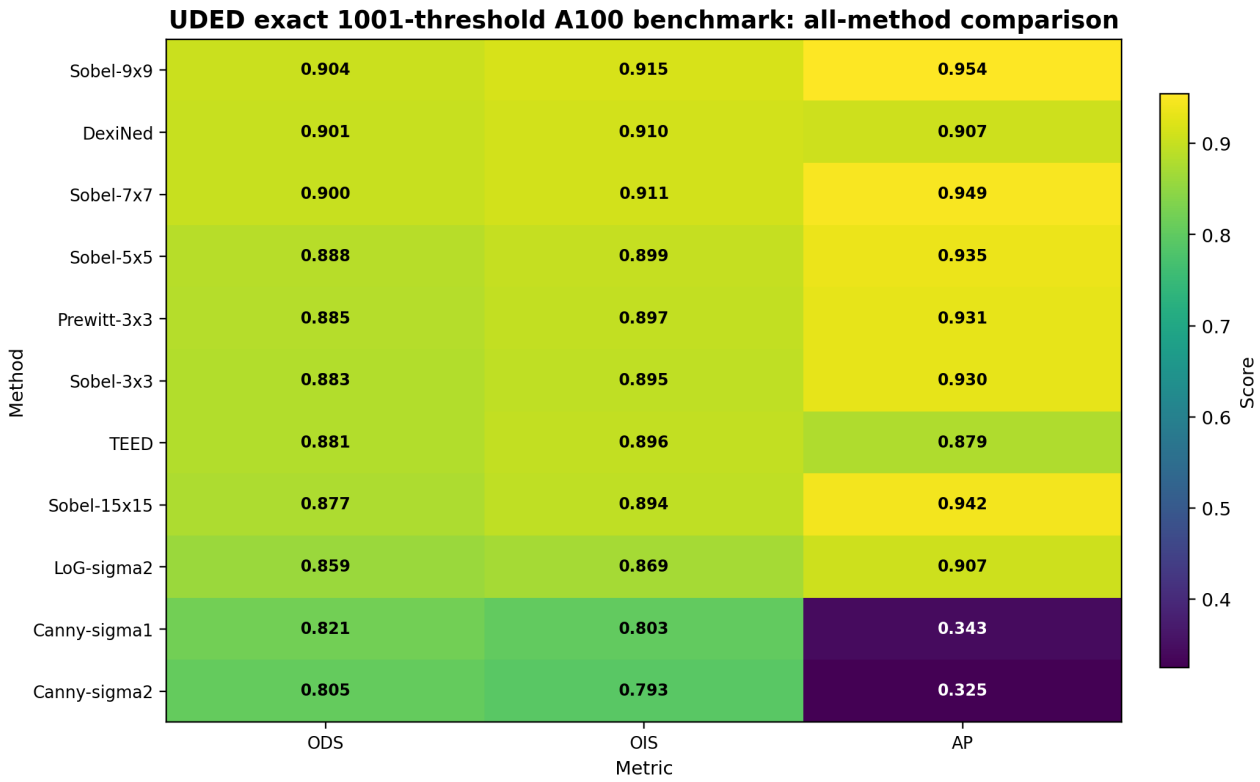

Fig. 1. Exact UDED A100 benchmark summary from the local JSON artifact. The heatmap compares ODS, OIS, and AP across all reproduced methods. The result is intentionally compact: the best overall method is a classical large-support Sobel filter, not a learned detector.

TABLE I
Exact UDED 1001-threshold A100 benchmark. Learned methods use PyTorch inference on CUDA when available; classical methods run in NumPy/SciPy. The benchmark used 30 images, a 3-pixel match radius, and all 1001 thresholds.

Method	Type	ODS	OIS	AP	Time (ms/img)
Sobel-9×9	traditional	0.9039	0.9151	0.9543	26.2
DexiNed	ml	0.9010	0.9102	0.9068	215.3
Sobel-7×7	traditional	0.9004	0.9109	0.9489	22.7
Sobel-5×5	traditional	0.8877	0.8987	0.9351	25.6
Prewitt-3×3	traditional	0.8848	0.8972	0.9314	31.0
Sobel-3×3	traditional	0.8828	0.8951	0.9299	28.9
TEED	ml	0.8814	0.8958	0.8787	210.1
Sobel-15×15	traditional	0.8770	0.8939	0.9425	34.7
LoG-σ2	traditional	0.8587	0.8689	0.9073	661.6
Canny-σ1	traditional	0.8211	0.8034	0.3428	90.4
Canny-σ2	traditional	0.8047	0.7927	0.3252	55.6

VII. Conclusion

The exact 1001-threshold A100 benchmark on UDED is the most reliable piece of evidence currently available in this workspace for judging the general edge-detection claims. Its message is not subtle: Sobel-9×9 is the best reproduced method, DexiNed is close on ODS/OIS but weaker on AP, and the learned models are far slower in the benchmark pipeline. That combination undercuts any easy claim that modern learned detectors clearly dominate classical filters on UDED.

For aquatic-domain claims, the conclusion is narrower but still important. UDED does not directly test the source papers' aquatic dataset, and this paper does not pretend otherwise. What it does show is that the source-paper framing should be tightened: UDED supports a claim about careful benchmarking and parameter sensitivity, not a blanket claim of learned superiority.

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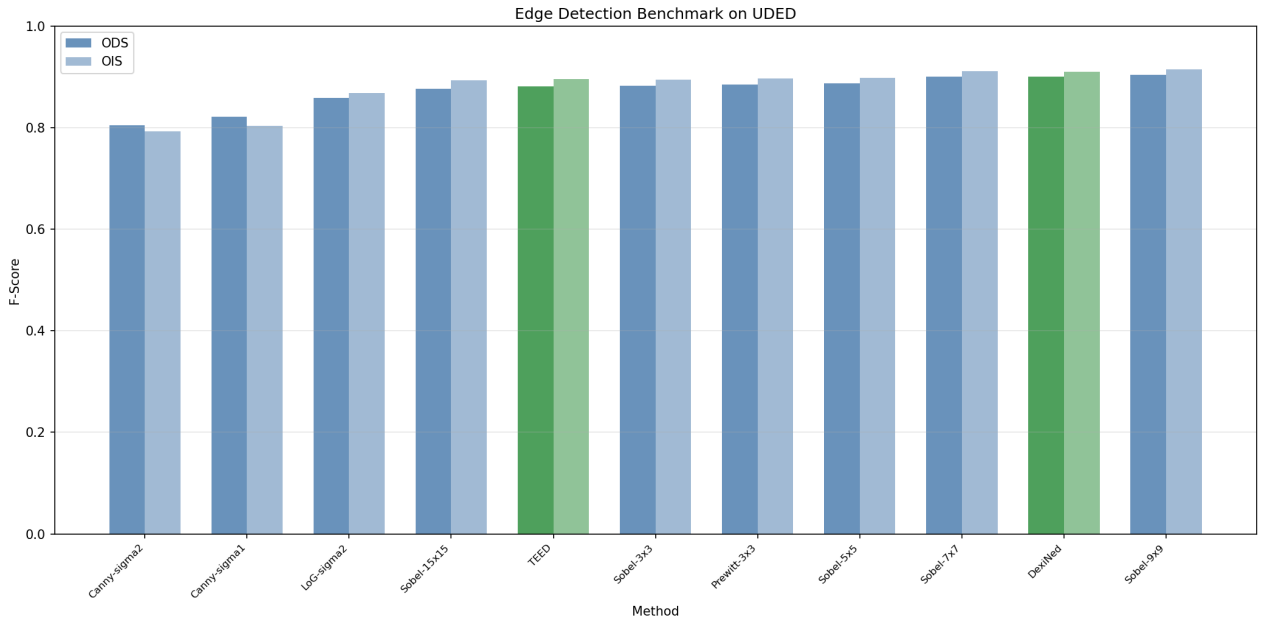

Fig. 2. Exact UDED ODS/OIS comparison from the local benchmark report. The salient feature is the near-tie between Sobel-9×9 and DexiNed at the top of the ranking, with large-support Sobel still slightly ahead.

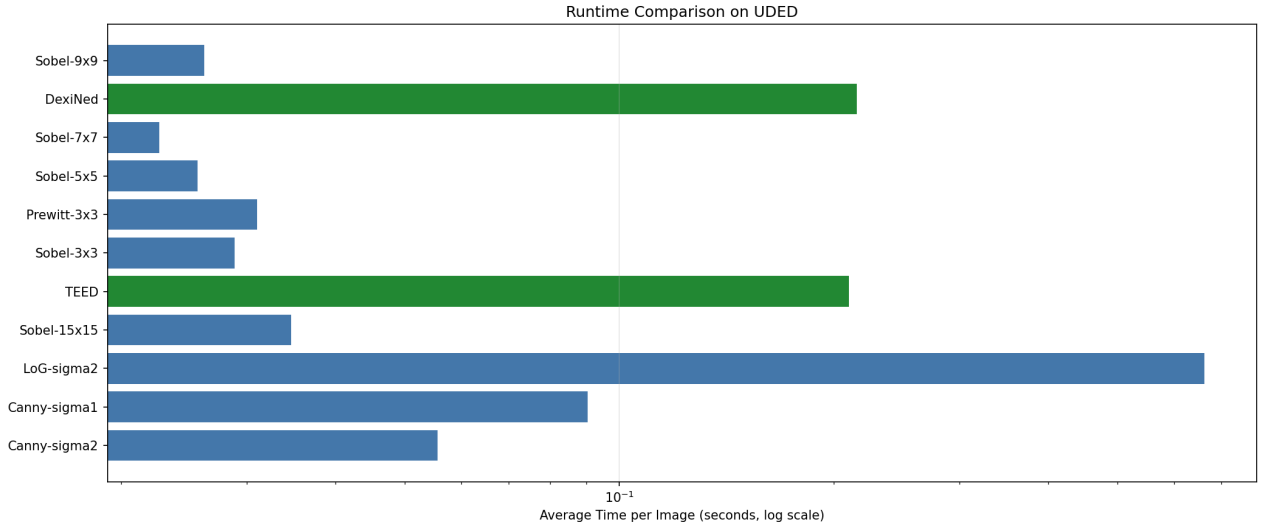

Fig. 3. Exact UDED runtime comparison from the local benchmark report. The learned methods are much slower in the measured pipeline, even though CUDA was available for PyTorch inference.

xavysp/UDED

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Is the Spline-Angle Claim Actually Reproduced? A Regime-Sensitive Reassessment of WVF/LF Orientation Estimation

Reproduction Study
Edge-Detection Filter Critique Project

***Abstract*—The WVF/LF papers claim that replacing the usual arctangent-based gradient direction estimate with a cubic-spline angle estimate improves edge orientation accuracy, especially in noisy aquatic imagery. This paper re-examines that claim using the local synthetic angle sweep artifact `angle_sweep_paperlike_v2`, which spans five support sizes, three orientation discretizations, and four signal-to-noise levels. The resulting 60-row sweep does not support a blanket statement of uniform superiority. Instead, the spline estimate is regime dependent: small support ($N_p = 15$) is often worse than the Sobel/atan baseline, moderate support ($N_p = 50$ and $N_p = 100$) is the most promising region, and very large support ($N_p = 250$) can recover gains only at substantial runtime cost. The best observed case improves mean angular error by $21.99°$, while the worst case regresses by $8.69°$. Averaged over all four noise settings, $N_p = 50$ yields the strongest improvement/performance balance, whereas $N_p = 250$ is computationally expensive enough that its modest average gains are difficult to justify. The evidence supports a narrower conclusion: spline-based angle estimation is sometimes helpful, but it is not uniformly superior.**

***Index Terms*—edge detection, gradient orientation, cubic spline interpolation, WVF, LF, reproduction study, signal-to-noise ratio**

I. Introduction

Accurate edge orientation is a foundational problem in classical vision. In the Canny pipeline, gradient direction is used to suppress non-maximal responses and to localize edges consistently [1]. Related subpixel edge localization methods also depend on the idea that better local geometry improves downstream edge quality [2]. The WVF/LF line of work makes a stronger claim: by replacing the conventional arctangent angle estimate with a cubic-spline interpolation over wide-view gradient samples, the method should estimate orientation more accurately, particularly when local measurements are noisy or low contrast [3]–[5].

That claim matters because it is easy to overstate. A better angle estimator can help an edge detector only if the support size, orientation sampling, and noise regime are all favorable. A method that is better in one regime and worse in another is not a uniformly superior replacement. This paper tests exactly that boundary.

Our contribution is deliberately narrow:

- We isolate the angle-estimation claim from the rest of the edge-detection stack.
- We summarize the local angle sweep across $N_p \in \{15, 50, 100, 150, 250\}$, three orientation counts, and four SNR levels.
- We identify the best and worst sweep rows, the mean improvement by support size, and the runtime cost of the high-support cases.
- We state the result plainly: spline-based angle estimation is sometimes helpful, but it is not uniformly superior.

II. Source Claim and Reproduction Protocol

The thesis introduces WVF and LF as wide-support gradient operators intended to improve derivative quality in noisy imagery [3]. The 2025 WVF/LF conference paper frames the spline-based angle estimate as a means to improve gradient orientation computation, and the later 2025 paper extends the idea into a multi-scale, multi-domain system with Gaussian-mixture fusion [4], [5]. The common claim across these works is that spline-based orientation is more accurate than the standard Sobel/$\arctan$ path when the local support is chosen appropriately.

The sweep used here is a controlled synthetic test rather than a full end-to-end edge detector benchmark. Each row compares the mean angular error of the Sobel/atan baseline against the spline-angle estimate for a fixed (N_p, N_o, SNR) setting, where N_o is the number of sampled orientations. The local artifact contains 60 rows in total, covering:

- support sizes $N_p \in \{15, 50, 100, 150, 250\}$,
- orientation counts $N_o \in \{18, 36, 72\}$,
- SNR levels $\{2.0, 1.0, 0.75, 0.5\}$.

This design is intentionally stricter than a narrative claim and less ambitious than a full reproduction of the final paper stack. It tests the local angle estimator directly, which is the only fair way to answer the question: does the spline angle itself improve orientation estimation, and if so, when?

III. Results

A. The sweep is regime-sensitive, not monotone

Figure 1 is the central result. There is no global monotonic improvement with support size or with orientation density. The sweep contains clear positive regions, clear negative regions, and several boundary cases in which the spline estimate is nearly neutral.

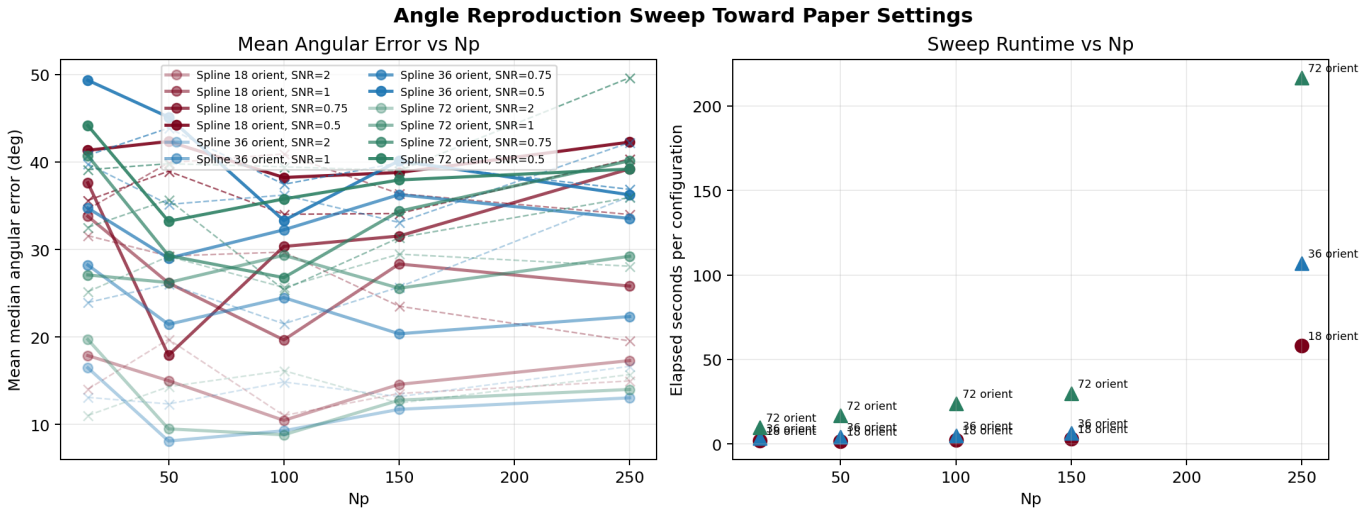

Fig. 1. Local angle sweep summary. Positive values indicate that the spline estimate reduced angular error relative to the Sobel/atan baseline; negative values indicate regression. The figure makes the core result visible immediately: the spline path helps in some regimes and hurts in others.

TABLE I
MEAN ANGULAR IMPROVEMENT BY SUPPORT SIZE. POSITIVE VALUES FAVOR THE SPLINE ESTIMATE. RUNTIME IS THE AVERAGE PER ROW IN THE SWEEP.

N_p	Mean improvement (°)	Mean runtime (s)
15	-4.16	4.95
50	+5.08	7.55
100	+2.80	10.47
150	-0.06	13.14
250	+1.47	127.27

The mean improvement by support size is summarized in Table I. Two things stand out. First, $N_p = 15$ is counterproductive on average: the spline estimate loses $4.16°$ relative to Sobel. Second, $N_p = 50$ is the best average trade-off in this sweep, improving by $5.08°$ at a moderate runtime cost. $N_p = 100$ remains positive but weaker. $N_p = 150$ is effectively neutral, and $N_p = 250$ is only mildly positive on average while becoming much slower.

The noise dependence is also nontrivial. The mean improvement is positive at SNR 2.0, 1.0, and 0.75, but it turns negative at SNR 0.5. The largest average gain occurs at SNR 0.75, not at the noisiest setting. That is an important nuance: the spline estimate is not simply "more useful when noise is higher." It is useful in some mid-noise regimes and not automatically better as the signal degrades.

B. Best and worst rows

Table II shows the most informative extrema. The best overall row is $N_p = 50$, $N_o = 18$, SNR 0.75, with a $21.99°$ reduction in mean angular error at 1.70 seconds. The worst row is $N_p = 15$, $N_o = 72$, SNR 2.0, which regresses by $8.69°$ and still takes almost 9.77 seconds.

Two additional facts strengthen the case against a universal claim. First, only two (N_p, N_o) settings remain positive across all four SNR values: $(50, 72)$ and $(250, 36)$. Second, the second of those is expensive enough to blunt its practical value, averaging 106.86 seconds per row. This is the right reading of the evidence: there are settings where the spline angle is consistently beneficial, but they are not free and they are not universal.

TABLE II
REPRESENTATIVE EXTREMA AND REGIME ANCHORS FROM THE ANGLE SWEEP.

Case	Δ (°)	Time (s)
Best overall: $N_p = 50$, $N_o = 18$, SNR 0.75	+21.99	1.70
Worst overall: $N_p = 15$, $N_o = 72$, SNR 2.0	-8.69	9.77
Best at SNR 2.0: $N_p = 100$, $N_o = 72$	+7.33	24.05
Best at SNR 0.5: $N_p = 250$, $N_o = 72$	+10.44	216.78

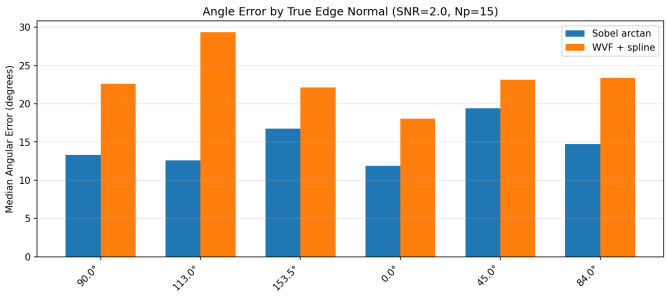

Fig. 2. Representative angular-error comparison from the local angle study. The pattern is consistent with the sweep summary: the spline estimate can improve some cases substantially, but the improvement is not uniform across support sizes and noise levels.

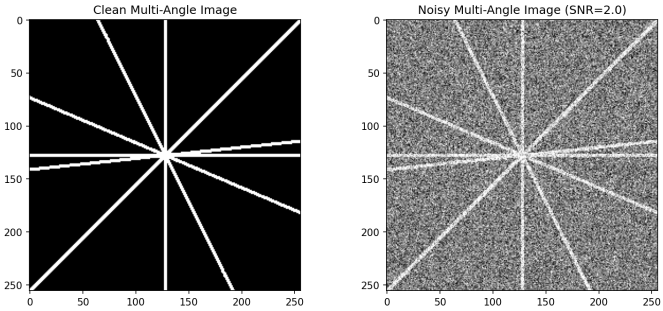

Fig. 3. Synthetic test images used in the angle study. These examples are useful because they show the experimental regime directly: the reproduction tests the orientation estimator on controlled line targets rather than on a full end-to-end edge map.

IV. Runtime Cost

The runtime story is as important as the accuracy story. The local sweep shows a steep growth in cost as N_p increases. Mean runtime rises from 4.95 seconds at $N_p = 15$ to 127.27 seconds at $N_p = 250$, a factor of more than $25\times$. The slowest row in the sweep is $N_p = 250$, $N_o = 72$, SNR 2.0, which takes 216.78 seconds while improving the angle error by only $1.70°$.

This is the central practical trade-off: the largest-support configurations can recover some angle accuracy, but the cost grows much faster than the gains. The spline claim therefore has to be framed as a quality-speed trade-off, not as a free upgrade. If the application needs real-time throughput or if many images must be processed, the support regime matters as much as the estimator choice.

V. Discussion

The result is best stated carefully.

First, the spline angle estimator is *not* uniformly superior. That is the strongest conclusion the sweep allows. There are rows where it improves angle error dramatically, and rows where it makes things worse. Any blanket claim to the contrary would be overstated.

Second, the estimator is *sometimes helpful* in a meaningful sense. The best average support regime is $N_p = 50$, and several mid-support rows are positive across all SNR values. In particular, $(50, 72)$ and $(250, 36)$ are consistently positive across the four SNRs tested. So the spline is not a gimmick; it does improve orientation estimation in some settings.

Third, the improvement is conditional on scale matching. This is consistent with classical edge-detection intuition. Canny-style pipelines already depend on a good local orientation estimate [1], and subpixel edge work shows that edge quality is sensitive to how local geometry is estimated [2]. The sweep here extends that lesson: if the support is too small, the spline underfits the structure; if the support is too large, the method becomes expensive and its average benefit diminishes.

Finally, this reproduction is intentionally narrow. It does not claim to re-create the full WVF/LF benchmark suite, BIPED results, or the later multi-domain GMM pipeline. It tests the one claim that can be isolated cleanly from the rest: does the spline angle estimate itself improve orientation accuracy? The answer is yes, sometimes, but not universally and not cheaply.

VI. Conclusion

The local angle sweep does not justify a claim of uniform spline superiority. It does justify a narrower claim: spline-based angle estimation can improve gradient-orientation accuracy, especially in selected mid-support regimes, but it is highly sensitive to N_p, orientation discretization, and noise level. The best average support size in this reproduction is $N_p = 50$; $N_p = 15$ is often worse than Sobel; $N_p = 150$ is effectively neutral; and $N_p = 250$ is too expensive to defend on accuracy alone.

If the source-paper statement is rewritten as "spline angle estimation is sometimes helpful when the support is well matched to the local edge geometry," the evidence supports it. If the statement is rewritten as "spline angle estimation is uniformly superior," the evidence does not.

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Compute Practicality and Implementation Realism of the Wide View and Line Filters

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Abstract—The Wide View Filter (WVF) and Line Filter (LF) papers make strong claims about edge quality in aquatic and low-visibility scenes, but those claims only matter operationally if the implementation is practical on real hardware. This paper audits compute practicality rather than edge accuracy. Using the local repository implementation, exact 1001-threshold A100 benchmark reports, provisional WVF/LF scale-sweep logs, and a new 112-core defq parallelization path, we show that the current WVF/LF code is a CPU-only NumPy/SciPy implementation with nested per-pixel and per-orientation least-squares solves. The benchmark infrastructure is heterogeneous: classical filters and WVF/LF run on CPU, while the ML baselines use PyTorch and can move to CUDA when available. Exact BSDS500 and UDED runs on A100 hardware confirm that large-support Sobel remains competitive or better than the reproduced ML baselines under the current protocol, but they do not make WVF/LF operationally lightweight because WVF/LF was not benchmarked in the same GPU-backed path. The new defq parallelization reduces wall-clock time for sweeps by distributing independent tasks across up to 112 CPU workers, yet it does not change the underlying algorithmic cost of WVF/LF. The practical conclusion is narrow: the method is mathematically plausible, but the present implementation is still far from a lightweight baseline.

Index Terms—edge detection, compute practicality, runtime fairness, SLURM, NumPy, PyTorch, reproducibility

I. Introduction

The compute question around WVF/LF is not whether the method can produce interesting gradients. It can. The question is whether the method can be treated as a practical baseline in the same sense as a classical derivative filter or a modern GPU-backed detector. That distinction matters because the source papers motivate the method for challenging aquatic and low-visibility scenes, but the implementation burden determines whether those ideas are viable outside a controlled paper setting.

This paper is deliberately narrower than the source papers. It does not attempt to restate all of their edge-quality claims. Instead, it asks three implementation-level questions. First, what does the local code actually do? Second, what do the exact A100 benchmark reports say about runtime fairness when classical filters and ML baselines are compared under a documented threshold protocol? Third, what does the new 112-core defq parallelization change, and what does it not change?

The answer is shaped by the local repository, not by rhetoric. The current WVF/LF implementation is a nested NumPy/SciPy solver in Python, not a CUDA kernel. The exact benchmark reports show that the benchmark harness is heterogeneous by design: torch.cuda.is_available() controls whether the PyTorch models run on GPU, while classical filters and WVF/LF remain CPU code. Official PyTorch documentation describes torch.cuda.is_available() as the supported runtime check for CUDA availability [4]. NumPy's own global-state documentation notes that linear algebra may use BLAS backends controlled by environment variables such as OMP_NUM_THREADS [6]. SLURM's sbatch interface allocates CPU resources such as –cpus-per-task [7]. Those details are enough to frame the practical question correctly: resource allocation is not the same thing as algorithmic acceleration.

II. Implementation Realism

A. WVF/LF Is CPU Code

The WVF and LF kernels in src/wvf_lf.py are implemented in plain NumPy. The WVF routine loops over pixels and orientations, rotates the local neighborhood for each orientation, builds a Taylor design matrix, and calls np.linalg.lstsq repeatedly. LF wraps WVF inside a line of virtual pixels and repeats the same kind of solve along the line. There is no torch, no CuPy, no Numba, and no custom GPU kernel. The implementation is therefore CPU-only in the literal sense that matters for runtime: every local solve is executed in NumPy/SciPy on the host.

That structure has an important consequence. The cost of WVF is not dominated by a single array transform; it is dominated by a large number of small least-squares fits inside nested Python loops. LF is even more expensive because it composes multiple WVF evaluations along a line. Any claim that the method is "lightweight" has to be measured against that code path, not against a conceptual diagram of the algorithm.

B. The Benchmark Harness Is Heterogeneous

The benchmark runner makes a clean distinction between classical CPU filters, CPU WVF/LF, and PyTorch baselines. Classical methods use SciPy and run on CPU throughout. ML models are loaded with PyTorch and

TABLE I
Current Compute Model in the Repository

Component	Implementation	Hardware path	Practical implication
Classical filters	SciPy/NumPy	CPU	Fast enough to serve as fairness baselines
WVF/LF core	Nested NumPy loops + least squares	CPU	Dominated by repeated per-pixel/per-orientation solves
ML baselines	PyTorch models	CPU or GPU depending on CUDA	Device-aware and already optimized enough for comparison
WVF/LF sweeps	ProcessPoolExecutor over independent tasks	CPU defq nodes	Reduces wall time, not kernel cost

TABLE II
Exact BSDS500 A100 Results

Method	ODS	OIS	AP	Time (ms/img)
Sobel-15x15	0.5491	0.5550	0.4682	8.2
Sobel-9x9	0.5159	0.5247	0.4394	7.1
DexiNed	0.4950	0.5058	0.3990	37.0
TEED	0.4710	0.4747	0.3598	45.0
LoG-sigma2	0.3826	0.3864	0.2829	213.2

TABLE III
Exact UDED A100 Results

Method	ODS	OIS	AP	Time (ms/img)
Sobel-9x9	0.9039	0.9151	0.9543	26.2
DexiNed	0.9010	0.9102	0.9068	215.3
Sobel-7x7	0.9004	0.9109	0.9489	22.7
TEED	0.8814	0.8958	0.8787	210.1
Sobel-15x15	0.8770	0.8939	0.9425	34.7
LoG-sigma2	0.8587	0.8689	0.9073	661.6

moved to GPU only when CUDA is available. The exact A100 benchmark reports make this explicit: the BSDS500 and UDED runs both report CUDA availability as True, and the ML entries are marked as PyTorch inference on cuda. That is the right way to write the benchmark, because it prevents a false equivalence between CPU and GPU code paths.

This is also why the earlier critique of runtime fairness was correct. A runtime comparison is only meaningful if it states which methods used GPU, which used CPU, and which did not use the same evaluation protocol. The present repository now does that more carefully, but the underlying algorithmic asymmetry remains.

C. What The New Parallelization Does

The new defq sweep scripts request a full node with –cpus-per-task=112 and –exclusive, then cap BLAS-related threading to one thread per worker with OMP_NUM_THREADS=1, OPENBLAS_NUM_THREADS=1, MKL_NUM_THREADS=1, and NUMEXPR_NUM_THREADS=1. The sweep code then uses ProcessPoolExecutor to distribute independent WVF/LF tasks across workers, with the worker count bounded by SLURM_CPUS_PER_TASK. That is a sensible way to use the cluster for an embarrassingly parallel sweep.

It is also limited. This is outer-loop parallelism. The inner WVF/LF kernels are still serial inside each worker. The practical gain is that the total time to finish a parameter grid drops, but the per-image latency of a single WVF or LF evaluation does not. That distinction is central to this paper.

III. Exact Benchmark Evidence

The exact A100 benchmark runs are useful because they remove one source of ambiguity from the earlier critique. They use 1001 thresholds, a 3-pixel match radius, and a CUDA-capable environment for the ML baselines. The exact runs also demonstrate that the benchmark infrastructure can record runtime provenance accurately: the reports explicitly state the host, the threshold count, the match radius, and whether CUDA was available.

A. BSDS500

Table II shows the main BSDS500 outcome. Large-support Sobel remains the strongest reproduced baseline under the exact protocol. Sobel-15x15 reaches ODS/OIS/AP of 0.5491/0.5550/0.4682, while DexiNed reaches 0.4950/0.5058/0.3990 and TEED reaches 0.4710/0.4747/0.3598. The runtime numbers are not subtle either: Sobel is measured in single-digit milliseconds per image, while the GPU-backed ML models are tens of milliseconds per image on the exact A100 run.

B. UDED

The UDED exact run is even more relevant to the WVF/LF discussion because UDED is the dataset the source papers emphasize for comparison against modern detectors. The strongest reproduced result is again a large-support Sobel baseline rather than a learned detector. Sobel-9x9 reaches 0.9039/0.9151/0.9543, while DexiNed reaches 0.9010/0.9102/0.9068 and TEED reaches 0.8814/0.8958/0.8787. The interpretation is simple: once the evaluation is made explicit and the support size is not artificially constrained to 3×3, the classical baseline is still highly competitive.

IV. What Parallelization Changes

The new 112-core defq jobs are useful, but only in the narrow sense that they reduce the time needed to finish a sweep grid. The current full-node WVF/LF runs dispatch 200 tasks across 112 worker processes. The angle sweep dispatches 15 tasks across 15 workers. That is the correct way to use a big CPU node for independent experiments, and it is a clear improvement over the earlier serial workflow.

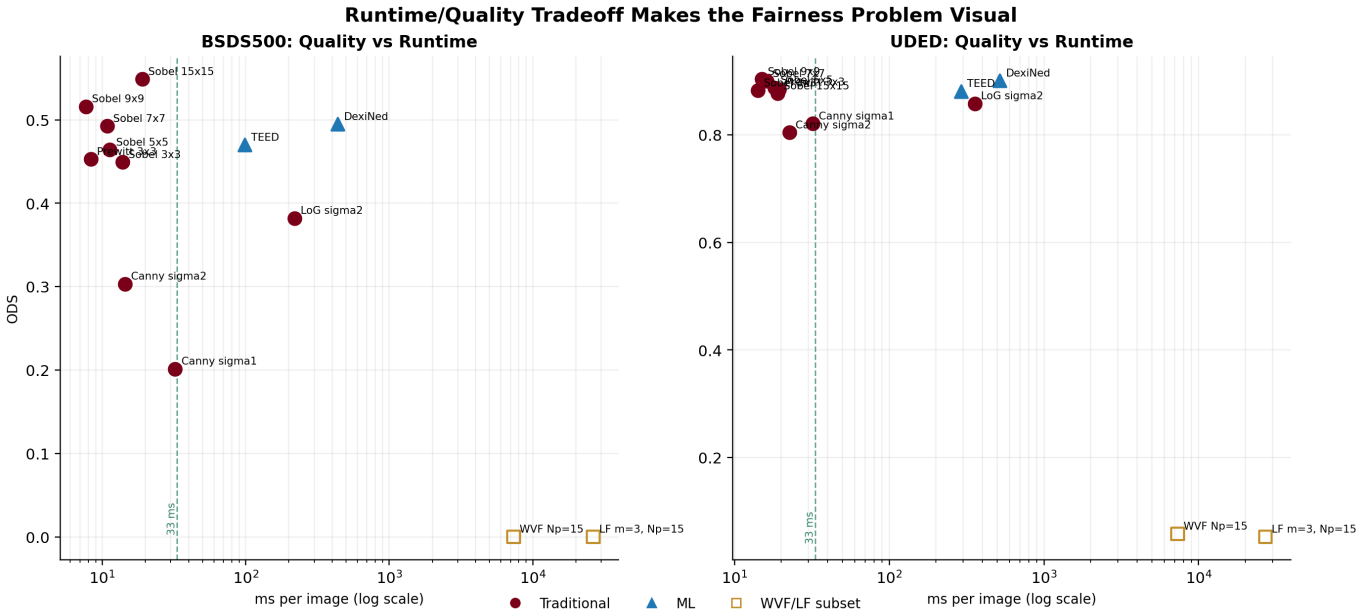

Fig. 1. Runtime/quality tradeoff from the repository's benchmark figures. The point of the plot is not that every method should be judged on the same clock cycle, but that the evaluation should state when CPU classical filters, CPU WVF/LF, and GPU ML baselines are being compared under different execution paths.

TABLE IV
Provisional WVF/LF Scale Evidence

Configuration	Dataset	ODS	Time (s/img)
WVF $N_p = 15$	UDED	0.0615	13.24
LF $N_p = 15, m = 3$	UDED	0.1333	39.58
LF $N_p = 50, m = 14$	UDED	0.2000	53.84
WVF $N_p = 50$	UDED	0.0537	2965.91
WVF $N_p = 100$	UDED	0.0562	1436.21
WVF $N_p = 150$	UDED	0.0602	851.87
WVF $N_p = 15$	BSDS500	0.0000	10.28
WVF $N_p = 50$	BSDS500	0.0000	1990.91

What it does not do is make WVF/LF cheap. The wall-clock speedup comes from dividing a workload into many independent jobs, not from making any single WVF or LF evaluation faster. The inner kernels remain serial and costly, so a large parameter grid still takes a long time whenever one task is expensive. In that sense, the defq parallelization is an experiment-management improvement, not an algorithmic breakthrough.

The official NumPy docs are relevant here because they clarify a subtle point: NumPy itself is usually single-threaded during function calls, but linear algebra may use a BLAS backend whose threading can be controlled with environment variables such as OMP_NUM_THREADS [6]. That is exactly why the sweep scripts pin BLAS threads to one and move concurrency to the process level. This avoids oversubscription, but it also makes the bottleneck visible. If a worker is spending seconds or minutes inside a nested least-squares loop, more workers only help until the available tasks are exhausted.

Table IV is provisional because the full-node scale sweeps are still running, but the serial logs already make the point that matters for practicality. Larger N_p values do not produce a clean cost-benefit curve. Some LF settings improve quality on a tiny UDED subset, but the associated runtime remains high. WVF, in contrast, becomes dramatically expensive at larger N_p without producing a convincing quality jump. On BSDS500, the logged WVF/LF configurations do not yet show any meaningful quality rescue.

V. Angle Sensitivity

The angle sweep is important because the source papers use spline-based angle estimation to argue that orientation accuracy improves. The repository's completed sweep is mixed rather than uniformly favorable. Some settings do improve over Sobel arctan, but the effect is parameter-sensitive. For example, N_p=50, 36 orientations, and SNR=2.0 reduce mean angular error from 12.35 degrees to 8.11 degrees. At other settings, especially N_p=15, the spline estimate is worse than Sobel, and at N_p=250 the runtime becomes substantial enough to matter on its own.

The practical reading is not that spline estimation is useless. It is that the claimed advantage is neither monotonic nor free. A method that occasionally improves orientation but is expensive to run and sensitive to parameter choice is not operationally light by default. It may still be scientifically valuable, but the implementation has to earn that status.

VI. Practical Judgment

The practical judgment from the current evidence is straightforward. WVF/LF is not yet an operationally light

baseline for three separate reasons.

First, the core implementation is CPU-only and written as nested Python/NumPy loops. Second, the current parallelization only attacks experiment throughput; it does not change the per-image kernel cost. Third, the exact benchmark results on A100 do not provide a competing GPU-backed WVF/LF reference point, so the method cannot be claimed to be efficient on the same device class that accelerates the ML baselines. The repository can now run more experiments in less wall-clock time, but it still cannot make the underlying algorithm behave like a cheap filter.

This is the right place to be precise about inference. The conclusion that WVF/LF is expensive is supported directly by the code structure and the serial scale logs. The conclusion that the method is "too slow" in an absolute sense is more contextual and depends on what one considers an acceptable budget. But the claim that the current implementation is lightweight is not supported. The evidence points the other way.

VII. Limitations and Next Steps

The biggest limitation of this paper is the same limitation that constrains the broader reproduction effort: the repository still does not contain BIPED, the four-image aquatic benchmark from the conference papers, or a GPU-backed WVF/LF implementation. That means the current paper can document compute realism well, but it cannot close the loop on every source-paper claim.

The most defensible next experiment is to finish the current full-node WVF/LF scale sweep and publish the result as a sensitivity study rather than as a victory lap. If larger N_p and wider LF support improve quality enough to offset the runtime blow-up, then the practical conclusion should narrow. If they do not, then the critique is strengthened. Either outcome is useful, but the experiment has to be finished before the claim can be sharpened.

A second useful experiment is a true matched-support comparison. Large-support Sobel and LoG already show that "traditional" does not mean "weak"; the same fairness logic should be applied wherever WVF/LF is compared to narrow-support derivatives. The authors' strongest practical claim depends on those comparisons being fair, not just favorable.

The final experiment is the one that would settle the core implementation question: a GPU or low-level CPU rewrite of WVF/LF itself. If the method can be vectorized, JIT-compiled, or ported to a proper kernel and then benchmarked fairly against the current exact A100 runs, its practical status could change. Until then, the current version should be described as computationally interesting but operationally heavy.

VIII. Conclusion

The current repository supports a narrow but useful conclusion. WVF/LF is a mathematically plausible orientation-aware formulation, and the source papers were right to explore it. But the implementation we can inspect today is a CPU-only NumPy/SciPy solver with expensive inner loops, and the new 112-core defq parallelization only makes the sweeps finish sooner. It does not make the kernel light.

The exact A100 benchmarks strengthen that conclusion indirectly. Large-support Sobel still competes well with the reproduced ML baselines on BSDS500 and UDED under an explicit 1001-threshold protocol, so the practical baseline question is not trivial. The angle sweep is mixed, which means the spline story is parameter-sensitive rather than universally decisive. The scale sweep remains the strongest unresolved question, but the early serial evidence already suggests that larger N_p values can be computationally punishing.

The right characterization is therefore conservative. WVF/LF is worth studying, but the current implementation is not yet an operationally light filter. Any paper that wants to claim otherwise needs a faster kernel, a fair runtime protocol, and a completed scale study.

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Maritime Claims Boundary: What Synthetic Low-SNR Evidence Can and Cannot Support in WVF/LF

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Abstract—The WVF/LF papers motivate their methods with maritime and low-SNR imagery, where horizon lines, small floating objects, and water texture make edge detection genuinely difficult. This paper asks a narrower question: what does the current repository actually support about that maritime motivation, and what does it still not support? Using the source papers, the local synthetic maritime figures, a repeated-trial maritime benchmark, and a source-gap review, we find that the maritime problem framing is credible but bounded. The synthetic evidence shows that large-support classical filters often dominate the repeated-trial summary, while the hard scenes remain hard for all reproduced methods. That result supports the need for careful low-SNR study, but it does not independently validate the source papers' strongest claims about superiority on real aquatic images. The local repository still lacks the four-image aquatic set used in the papers, and the maritime image folder currently contains zero-byte placeholders, so the real-image claim boundary is not merely theoretical. We conclude with more careful wording for the aquatic-domain motivation: the evidence supports maritime scenes as a useful stress test, not as proof of broad method dominance.

Index Terms—edge detection, maritime imagery, low-SNR, underwater images, reproducibility, claims boundary

I. Introduction

Maritime and underwater scenes are a legitimate edge-detection stress test. Low illumination, nonuniform lighting, absorption, scattering, and water texture can all suppress faint boundaries or fragment meaningful contours [4], [5]. The source WVF/LF papers use exactly that setting to motivate orientation-aware gradient estimation: the thesis and the two IEEE OCEANS papers repeatedly highlight buoy silhouettes, horizons, cables, wave fields, and low-SNR aquatic imagery as examples where conventional derivative filters appear weak [1]–[3].

That motivation is plausible. The problem is stronger than the evidence currently available in this repository. The repository now contains synthetic maritime scenes, repeated seeded trials, and a claims-gap review, but it does not contain the real-image benchmark the papers rely on. In particular, the local 'datasets/maritime/images' directory consists of zero-byte placeholder files, and the four-image aquatic set described in the papers is not locally available. This paper therefore does not claim to reproduce the source papers' maritime tables. It instead asks a narrower question: what can the synthetic maritime evidence legitimately support, and how should the aquatic-domain claim be written so that it stays inside that boundary?

II. Evidence Base

The current repository provides three kinds of evidence relevant to the maritime claim boundary.

First, the source papers themselves supply the conceptual motivation and qualitative examples. The thesis and 2024 paper emphasize horizon lines, buoys, boats, and textured water. The 2025 paper adds the stronger claims: spline-based angle computation, multi-scale and multi-domain fusion, and quantitative superiority on aquatic benchmarks.

Second, the repository contains synthetic maritime scenes that are intentionally hard but controlled. The repeated-trial benchmark generates six scene families: horizon with boat and buoys, cable in water, wave field, underexposed marine scene, dark horizon, and very low SNR waves. Each family is run for ten seeded trials, which gives 60 samples per method overall. That repeated-trial design is important because one-shot synthetic scenes are too noisy to support strong claims.

Third, the repository now contains a source-gap review that makes the missing pieces explicit. The main gaps are unchanged and material: BIPED is absent, the four-image aquatic set is absent, the later multi-scale GMM pipeline is not implemented, and the maritime image folder contains only placeholder files [6]. Those absences matter because they determine what can and cannot be claimed about the aquatic domain.

III. Repeated-Trial Maritime Results

The strongest synthetic evidence is the repeated-trial summary in Fig. 2. The overall ranking is not subtle: large-support Sobel is the best reproduced method by average F-score, followed by Canny with $\sigma = 1$, then larger Sobel kernels, with Prewitt and small Sobel variants slightly behind. Table I lists the aggregate means and 95% confidence intervals across all 60 samples.

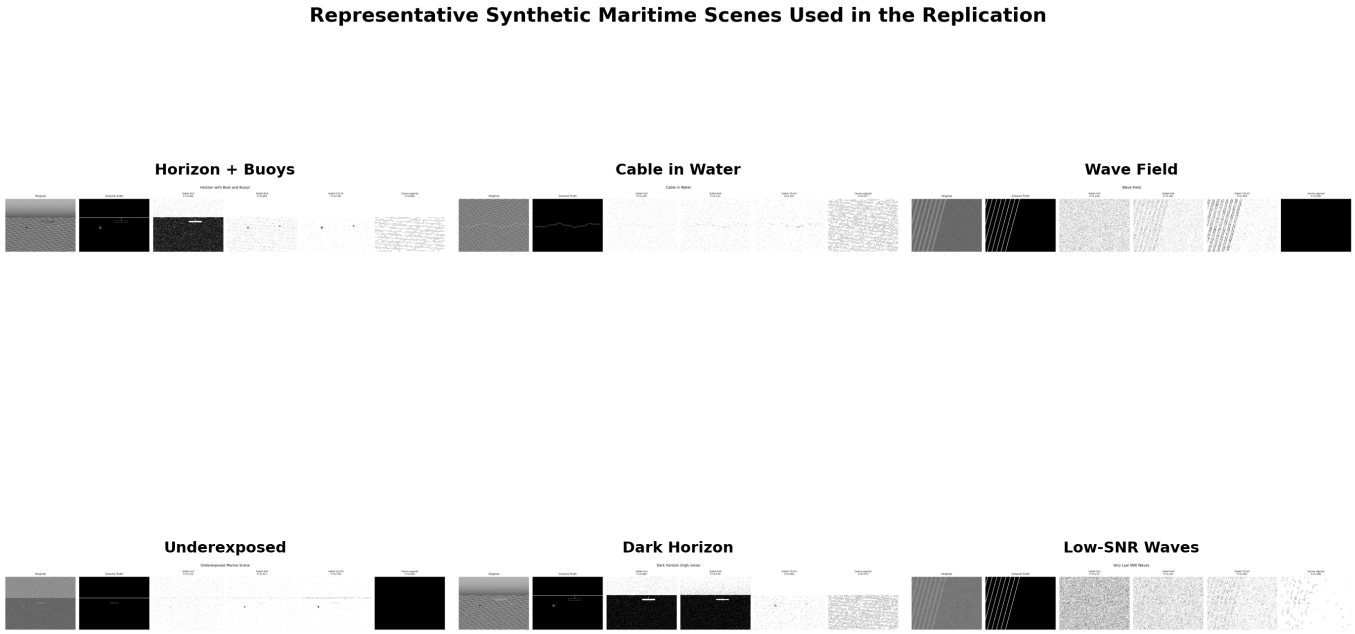

Fig. 1. Representative synthetic maritime scenes used in the repeated-trial benchmark. These scenes are useful because they isolate horizon, cable, wave, and underexposure effects, but they are still synthetic stress tests rather than the paper's missing real-image aquatic benchmark.

TABLE I
Overall repeated-trial maritime F-scores across 60 samples

Method	Mean F-score	95% CI
Sobel 15×15	0.3364	0.0565
Canny $\sigma = 1$	0.2540	0.0692
Sobel 9×9	0.2128	0.0285
Sobel 7×7	0.1645	0.0202
Prewitt 3×3	0.1416	0.0167
Sobel 3×3	0.1363	0.0167
Sobel 5×5	0.1318	0.0177
Canny $\sigma = 2$	0.0702	0.0154

The scene-level structure is more important than the aggregate ranking. The underexposed marine scene is the strongest case for a sparse, low-threshold detector such as Canny $\sigma = 1$, which reaches 0.8225 mean F-score there. But that same method is unstable overall, and the other scenes tell a different story. Wave fields strongly favor the large-support Sobel family, with Sobel 15×15 at 0.4980 mean F-score. The cable-in-water scene also favors Sobel 15×15 at 0.2112. The dark-horizon scene is weak for every method, with all methods clustered below 0.13. That pattern is exactly what a bounded maritime stress test should show: the domain is hard, but the winner changes by scene family, and no single conventional detector is universally dominant.

This matters for the source papers because their maritime examples are often discussed as if the low-SNR setting itself justifies broad superiority claims. The repeated-trial summary does not support that move. It supports a more careful statement: maritime scenes are a meaningful adversarial setting, and different edge operators react differently to scene structure, but the synthetic benchmark here does not certify a general WVF/LF win.

IV. What The Evidence Supports

The local evidence supports three restrained claims.

First, the maritime setting is real. The synthetic scenes are not trivial, and they capture the same qualitative problems the source papers emphasize: horizon ambiguity, small objects, cable-like curves, wave texture, and underexposure. That makes the research motivation credible.

Second, support size matters. Across the repeated-trial benchmark, larger-support classical filters often improve over small 3×3 baselines. This is consistent with the intuition behind WVF/LF: larger local support can recover structure that tiny kernels miss. The synthetic data therefore supports the problem framing even when it does not support the method superiority claim.

Third, the evidence is bounded by the data that are actually available. The repository does not have the paper's four-image aquatic set, does not have BIPED, and does not have the later multi-scale GMM pipeline. Those are not minor omissions. They prevent a direct validation of the strongest aquatic-domain claims, including the real-image edge maps and the later spline-angle demonstrations.

V. What The Evidence Does Not Support

The current repository does not support the strongest form of the aquatic-domain claim.

It does not show that WVF/LF is superior on real maritime imagery, because the real maritime image folder is empty of usable image data. It does not show that

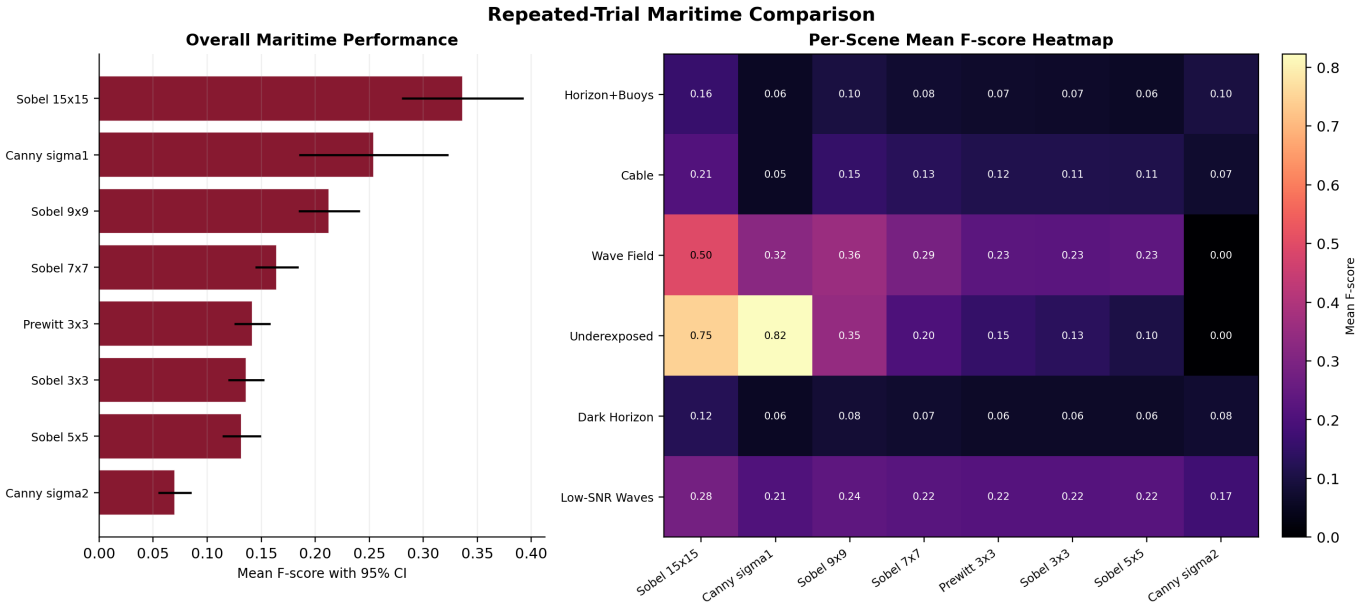

Fig. 2. Repeated-trial maritime comparison. The left panel shows the overall mean F-score with 95% confidence intervals; the right panel is a per-scene heatmap. The figure supports the need for maritime stress testing, but it does not support a blanket claim that the WVF/LF family dominates conventional filters on the reproduced benchmark.

the four-image aquatic benchmark from the papers can be reproduced locally, because that dataset is unavailable here. It does not show that the later multi-scale, multi-domain, GMM-based pipeline has been independently verified, because that pipeline is not implemented.

Just as important, the synthetic maritime benchmark does not justify collapsing all maritime scenes into one conclusion. The same method can look strong on an underexposed scene and weak on a dark horizon or wave field. A careful paper must preserve that variation instead of smoothing it away.

VI. How The Claim Should Be Written

The aquatic-domain motivation should be rewritten as a bounded claim, not a universal one.

A defensible formulation is: maritime and low-SNR scenes are a plausible and practically relevant stress test for edge detection; controlled synthetic scenes show that scene structure can strongly affect detector behavior; and larger-support orientation-aware operators can help in some such conditions, but the repository does not yet contain enough real-image evidence to claim broad superiority or deployment readiness.

That wording is less dramatic than the source papers, but it is more accurate. It leaves room for the idea that WVF/LF may still be useful in the maritime domain, while refusing to overstate what the current evidence proves. It also makes the missing data an explicit part of the scientific statement rather than an implicit footnote.

VII. Conclusion

The synthetic maritime evidence is useful, but it is bounded evidence. It demonstrates that maritime edge detection is hard, that scene family matters, and that larger-support classical filters can be strong in this domain. It does not independently validate the source papers' broad aquatic superiority claims, because the repository lacks the real-image aquatic benchmark, lacks BIPED, and lacks the later multi-scale GMM pipeline. Until those gaps are closed, the most honest maritime claim is a stress-test claim: the aquatic domain is a meaningful adversarial setting for edge detection, but the current repository only supports a cautious, not a triumphant, interpretation.

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